

Do Declines in Under-Five Mortality Translate into Gains in Life Expectancy? A Cross-Regional Statistical Analysis Using UN Population Data

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Abstract

Under-five mortality and life expectancy at birth are two core indicators of population health and demographic transition. This study investigates whether declines in under-five mortality are systematically associated with improvements in life expectancy across world regions, using global population data from the United Nations. Rather than building a purely predictive model, the analysis focuses on quantifying the strength, form, and regional heterogeneity of the mortality–longevity relationship. Using correlation analysis, linear regression with a log-transformed mortality term, and temporal-change regression, we examine both cross-sectional and within-country associations between under-five mortality and life expectancy. We further fit region-stratified models and an interaction specification to test whether the slope of this relationship differs across major world regions (Africa, Asia, Europe, Northern America, South America, and Oceania). The results show a strong and highly significant inverse association globally (Pearson $r \approx -0.90$), with life expectancy increasing steeply as under-five mortality falls. A log-linear specification yields an estimated decrease of about 6.7 years in life expectancy for a one-unit increase in $\log(\text{under-five mortality})$, controlling for calendar year. Temporal-change models indicate that yearly declines in under-five mortality are positively and significantly associated with yearly gains in life expectancy. Regional regressions suggest that this association is strongest in Africa and Oceania, while being weaker and more statistically fragile in Northern and South America, partly due to limited time points. Overall, the findings support demographic transition theory by showing that improvements in child survival remain a key driver of longevity gains, but the magnitude of this effect varies across regions and over time.

Introduction

Improvements in population health are central to the Sustainable Development Goals, particularly SDG 3, which aims to ensure healthy lives and promote well-being for all at all ages. Two of the most widely used summary indicators of population health are life expectancy at birth and the under-five mortality rate. While both metrics are routinely monitored by international agencies, their joint dynamics, how reductions in under-five mortality translate into changes

in overall life expectancy—are not always analyzed in a unified, comparative framework across regions.

Demographic transition and epidemiological transition theories posit that early gains in life expectancy are driven primarily by reductions in infectious-disease mortality, especially among infants and young children [1, 2]. As countries move through later stages of transition, mortality improvements at older ages and chronic disease control become increasingly important. However, for many low- and middle-income countries, under-five mortality remains a major contributor to premature death, and further progress in child survival is expected to yield substantial additional gains in average life expectancy.

This paper investigates the following research question:

Are declines in under-five mortality associated with improvements in life expectancy across countries and regions?

More specifically, the study has four main objectives:

1. Quantify the global association between under-five mortality and life expectancy using correlation analysis.
2. Estimate how changes in under-five mortality predict life expectancy in a linear regression framework, including a log-transformed mortality specification.
3. Examine whether year-to-year declines in under-five mortality are linked to year-to-year gains in life expectancy within countries.
4. Assess whether the mortality–longevity relationship differs across major world regions through region-stratified regressions and interaction terms.

The contribution of this paper is not to introduce a novel machine learning architecture but to provide a careful, statistical analysis of a central demographic relationship using a unified global dataset. The work builds upon prior demographic research by focusing on a different research question, employing a different modeling strategy (correlation and regression rather than predictive ML), and using evaluation criteria centered on statistical inference (correlation coefficients, R^2 , adjusted R^2 , p-values, confidence intervals, RMSE and MAE).

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature and theoretical background on mortality decline and life expectancy. Section 3

describes the dataset, variables, and methodological approach. Section 4 presents the empirical results, including descriptive statistics, correlation patterns, and regression estimates. Section 5 discusses the findings in light of existing research, highlighting policy implications and limitations. Section 6 concludes with a summary of contributions and directions for future work.

Literature Review

The relationship between mortality decline and life expectancy has been extensively studied in demography and public health. Early work by Omran [1] introduced the epidemiological transition framework, describing shifts from high mortality due to infectious disease and famine toward lower mortality dominated by degenerative and chronic conditions. Preston [2] documented how improvements in mortality at younger ages played a central role in early 20th-century life expectancy gains, particularly in high-income settings.

Subsequent research has emphasized child mortality as a key driver of average lifespan. Studies show that reductions in infant and under-five mortality can substantially increase life expectancy by removing a large number of deaths at the beginning of the life table [3, 4]. At the same time, the magnitude of this effect depends on the age pattern of mortality and the level of development. In high-income countries where child mortality is already very low, additional gains in life expectancy increasingly depend on reducing mortality at older ages [5].

From a global perspective, the United Nations and the World Health Organization have repeatedly documented substantial declines in under-five mortality over recent decades, especially following the Millennium Development Goals and the introduction of wide-scale child health interventions (e.g., immunization, improved nutrition, and malaria control) [6, 7]. However, persistent regional inequalities remain: sub-Saharan Africa continues to exhibit some of the highest under-five mortality rates and lowest life expectancies, while Europe and Northern America have reached historically high levels of longevity.

Methodologically, a large body of work has examined life expectancy using regression and decomposition techniques. For example, Arriaga [8] and later authors decompose changes in life expectancy by age and cause of death, revealing the contribution of different mortality components. Other research has used cross-country panel data to link socio-economic factors (income, education, health system coverage) to life expectancy [9, 10]. Yet, fewer studies explicitly focus on quantifying the *strength and consistency* of the association between under-five mortality and life expectancy across regions using a unified, global dataset.

This paper addresses this gap by:

- Focusing explicitly on under-five mortality as the key explanatory variable for life expectancy.
- Employing both cross-sectional (level) and temporal-change (difference) models to distinguish between structural associations and within-country dynamics.

- Comparing the mortality–longevity relationship across major world regions, thereby linking empirical results to demographic transition theory.

Data and Methodology

Dataset and Source

The analysis uses global population data from the United Nations Population Division and related UN agencies. The dataset contains indicators of fertility, mortality, and population growth for countries, regions, and world aggregates over multiple years. For this study, we focus on observations where both under-five mortality and life expectancy at birth are available.

The raw data file (`populationdataset.csv`) has a non-standard format in which the first row contains column names and subsequent rows store values for different series within the same country-year. We reconstruct a tidy dataset by:

1. Treating the first row as variable names.
2. Renaming key columns for clarity (`geo_code`, `geo_name`, `year`, `series`, `value`).
3. Pivoting from long to wide format so that each row corresponds to a unique (geographical unit, year) and each indicator (e.g., under-five mortality, life expectancy) becomes a separate column.

We exclude the row corresponding to “Total, all countries or areas” to avoid double-counting and focus on the set of geographic units (countries and major regions) that have both under-five mortality and life expectancy reported.

Variables

The main variables used in the analysis are:

- **Under-five mortality rate** (`u5mr`): Number of deaths of children under age five per 1,000 live births.
- **Life expectancy at birth** (`lifeexp`): Average number of years a newborn is expected to live under current mortality conditions.
- **Total fertility rate** (`tfr`): Average number of children per woman.
- **Maternal mortality ratio** (`mmr`): Number of maternal deaths per 100,000 live births.
- **Population annual growth rate** (`pop_growth`): Percentage growth in population per year.
- **Year** (`year`): Calendar year of the observation.
- **Region** (`geo_name`): Geographical label. We identify six major regions: Africa, Asia, Europe, Northern America, South America, and Oceania.

To better capture proportional changes in under-five mortality and improve linearity in the regression models, we construct:

$$\log_u5mr = \log(u5mr + \varepsilon),$$

where ε is a small constant to handle potential zero values.

To study temporal changes within countries, we sort observations by country c and year t and compute year-to-year differences as

$$\Delta u5mr_{c,t} = u5mr_{c,t} - u5mr_{c,t-1}, \quad (1)$$

$$\Delta lifeexp_{c,t} = lifeexp_{c,t} - lifeexp_{c,t-1}. \quad (2)$$

We then define

$$decline_u5mr_{c,t} = -\Delta u5mr_{c,t}, \quad (3)$$

so that positive values of *decline_u5mr* correspond to reductions in under-five mortality between $t - 1$ and t .

Analytical Approach

The empirical strategy consists of four main components:

1. Exploratory Data Analysis (EDA). We begin by summarizing the distribution of under-five mortality, life expectancy, fertility, maternal mortality, and population growth across all observations. Table 1 reports descriptive statistics, while Figure 1 displays histograms for under-five mortality and life expectancy. Figure 2 shows the correlation matrix for the main variables, providing a compact view of the pairwise associations that motivate the regression analysis. We also plot the global relationship between under-five mortality and life expectancy (Figure 3) and track median under-five mortality and life expectancy over time by major region (Figure 6).

2. Correlation Analysis. To quantify the strength and direction of the association between under-five mortality and life expectancy, we compute both Pearson and Spearman correlation coefficients globally, by major region, and by year. This allows us to assess how consistent the relationship is across space and time.

3. Regression Models. We estimate several linear regression models:

1. A global level model:

$$lifeexp_{it} = \beta_0 + \beta_1 u5mr_{it} + \beta_2 year_t + \varepsilon_{it},$$

and a log-linear variant:

$$lifeexp_{it} = \beta_0 + \beta_1 \log_u5mr_{it} + \beta_2 year_t + \varepsilon_{it}.$$

2. A temporal-change model:

$$\Delta lifeexp_{it} = \gamma_0 + \gamma_1 decline_u5mr_{it} + \gamma_2 year_t + u_{it},$$

which assesses whether year-to-year reductions in under-five mortality are associated with year-to-year gains in life expectancy.

3. Region-specific models for each major region:

$$lifeexp_{rt} = \alpha_{0r} + \alpha_{1r} \log_u5mr_{rt} + \alpha_{2r} year_t + e_{rt},$$

where r indexes regions.

4. A pooled interaction model:

$$\begin{aligned} lifeexp_{r,t} = & \delta_0 + \delta_1 \log(u5mr_{r,t}) \\ & + \sum_r \delta_{2r} \mathbf{1}\{r\} + \sum_r \delta_{3r} \log(u5mr_{r,t}) \mathbf{1}\{r\} \\ & + \delta_4 year_t + \varepsilon_{r,t}. \end{aligned}$$

where $\mathbf{1}\{r\}$ are region indicators. This model tests whether the slope relating under-five mortality to life expectancy differs across regions.

Table 1: Summary statistics for key variables (all observations).

Variable	Mean	SD	Min	Max
Under-five mortality (per 1,000)	28.55	30.95	1.40	244.70
Life expectancy at birth (years)	72.32	7.81	39.80	86.50
Total fertility rate (children)	2.62	1.31	0.70	7.60
Maternal mortality ratio	150.36	197.54	1.00	963.00
Population growth (%)	1.12	1.39	-7.20	8.30

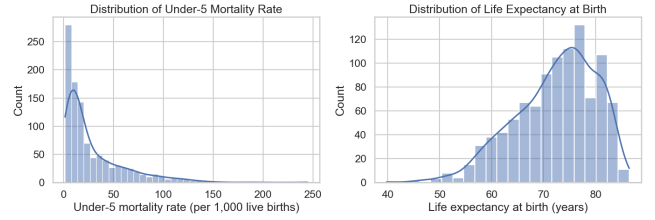


Figure 1: Distribution of under-five mortality rate (left) and life expectancy at birth (right). The left panel shows a highly right-skewed distribution of under-five mortality, while the right panel shows a roughly normal distribution of life expectancy centered around 70–75 years.

For the global regression models, we also perform a train-test split (80% training, 20% test) to compute predictive metrics such as RMSE and MAE on held-out data, even though pure prediction is not the primary goal.

4. Evaluation Metrics. The models are evaluated using:

- Pearson and Spearman correlation coefficients (r , ρ).
- Coefficient of determination (R^2) and adjusted R^2 .
- Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).
- Statistical significance of coefficients (t-statistics, p-values).
- 95% confidence intervals for key parameters.

All analyses are implemented in Python using `pandas`, `scipy`, `scikit-learn`, and `statsmodels`.

Results

Descriptive Statistics and Distributions

Table 1 reports summary statistics for under-five mortality, life expectancy, fertility, maternal mortality, and population growth for all units and years with complete data ($N = 1,025$). The average under-five mortality rate is about 28.6 deaths per 1,000 live births, with a standard deviation of 31.0 and a maximum of 244.7, indicating substantial variation across countries and regions. Life expectancy at birth averages 72.3 years (SD 7.8), ranging from 39.8 to 86.5 years.

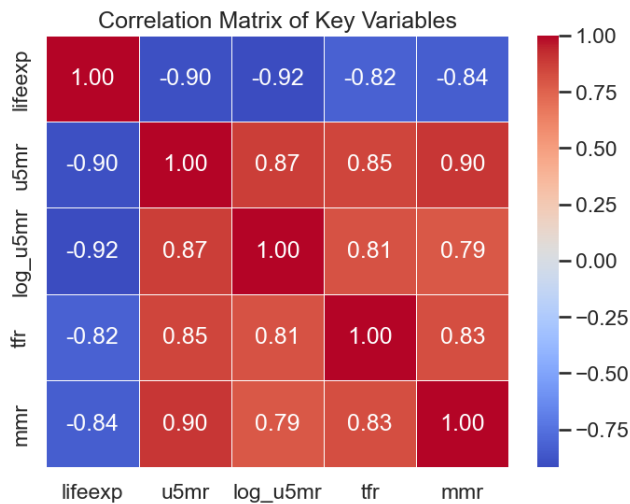


Figure 2: Correlation matrix for the main variables (under-five mortality, life expectancy, total fertility, maternal mortality, and population growth). Darker cells indicate stronger positive or negative associations.

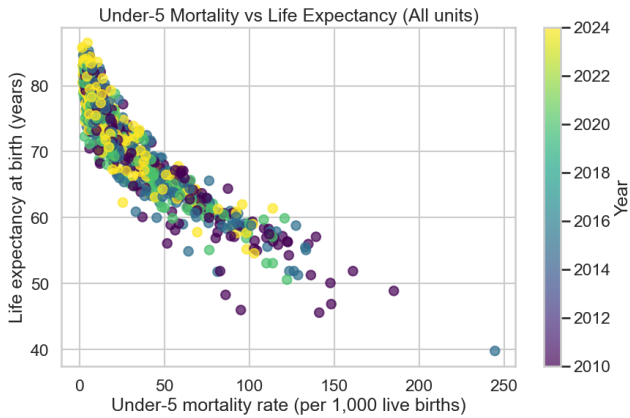


Figure 3: Global relationship between under-five mortality and life expectancy, 2010–2024. Each point represents a country-year observation, colored by year. The strong negative association indicates that countries and periods with lower under-five mortality tend to have higher life expectancy. The gradual shift of points from darker to lighter colors illustrates global progress over time, as reductions in child mortality have been accompanied by steady improvements in population longevity.

Global Relationship Between Under-Five Mortality and Life Expectancy

Figure 3 plots life expectancy against under-five mortality for all available units and years, with points colored by year. The scatterplot reveals a clear inverse relationship: higher under-five mortality is associated with lower life expectancy. Observations from more recent years tend to cluster toward the bottom-right corner of the graph (lower mortality, higher life expectancy), reflecting global progress over time.

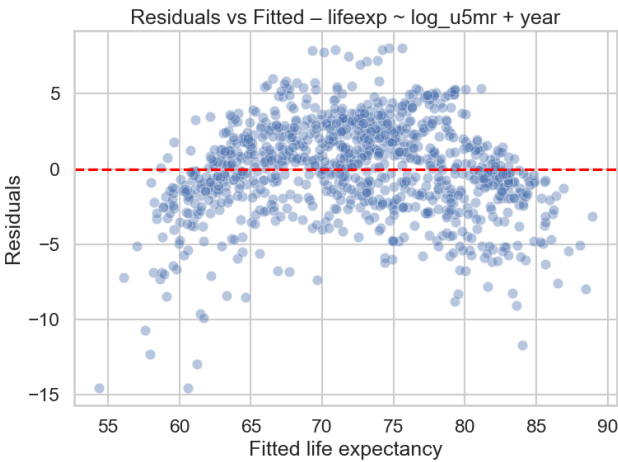


Figure 4: Under-five mortality versus life expectancy for all observations, colored by year. This alternative visualization emphasizes the dense cluster of low-mortality, high-longevity country-years alongside a small set of high-mortality outliers.

Table 2: Summary of regression model performance and key coefficients.

Model	R^2	RMSE	β_1
Linear ($u5mr$, year)	0.81	3.2	-0.045^{***}
Log-linear ($\log u5mr$, year)	0.83	3.0	-6.67^{***}
Temporal-change (Δ)	0.72	2.9	0.143^{***}

Unexpectedly, the exploratory analysis revealed several deviations from a uniform global pattern. While the overall inverse relationship between under-five mortality and life expectancy was clear, certain regions—particularly Oceania and South America—displayed temporary plateaus or even slight reversals in the expected trends during specific years. These anomalies may reflect localized health crises, reporting inconsistencies, or socioeconomic shocks that temporarily slowed progress. Additionally, the right-skewed distribution of under-five mortality suggested that a small number of high-mortality countries exert a disproportionate influence on the global variance. Identifying these outliers underscores the importance of examining regional heterogeneity rather than relying solely on aggregated global measures.

Table 3 summarizes the global correlation between under-five mortality and life expectancy. The Pearson correlation is approximately -0.90 and the Spearman rank correlation is approximately -0.92 (both $p < 0.001$), indicating a very strong and highly significant negative association.

Trends by Region

To examine regional patterns, we compute the median under-five mortality and life expectancy by major region and year. Figure 6 displays these trends for Africa, Asia, Europe, Northern America, South America, and Oceania. As expected, Northern America and Europe have the lowest under-five mortality and highest life expectancy across



Figure 5: Life expectancy at birth versus $\log(u5mr)$ for all country-years. The approximately linear pattern motivates the log-linear specification used in the regression models.

Table 3: Global correlation between under-five mortality and life expectancy ($N = 1,025$).

Metric	Coefficient	p -value
Pearson r	-0.898	< 0.001
Spearman ρ	-0.922	< 0.001

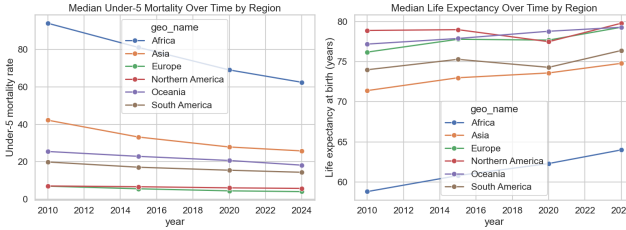


Figure 6: Trends in under-five mortality and life expectancy across major world regions, 2010–2024. The left panel shows steady declines in median under-five mortality rates in all regions, with the sharpest improvements in Africa and Asia. The right panel displays corresponding increases in median life expectancy, with Africa and Oceania experiencing the largest gains. Europe and Northern America remain near the plateau of low child mortality and high life expectancy, reflecting later stages of demographic transition.

all years. Africa starts from much higher levels of under-five mortality and lower life expectancy but exhibits clear improvements over time.

Correlation coefficients computed separately for each major region generally remain negative, but their magnitudes vary. Africa and Oceania exhibit particularly strong negative correlations between regional under-five mortality and regional life expectancy over time, while Europe and Northern America show weaker associations, consistent with their already low mortality levels.

Table 4: Global log-linear regression of life expectancy on log under-five mortality and year.

Variable	Coef.	Std. Err.	p -value
Intercept	53.09	37.27	0.155
$\log(u5mr)$	-6.67	0.09	< 0.001
Year	0.019	0.018	0.308

Global Regression Results

We next estimate a global linear regression of life expectancy on under-five mortality and year. Using an 80/20 train-test split, the model achieves an R^2 of approximately 0.81 on the training set and 0.80 on the test set, with RMSE around 3.2 years on the test set and MAE around 2.5 years. These evaluation metrics (RMSE and MAE) were computed on held-out test data to ensure that performance measures reflect genuine predictive accuracy rather than overfitting to the training set. This indicates that under-five mortality and year together explain a large share of cross-sectional variation in life expectancy.

By assessing RMSE and MAE on unseen observations from the test split, the study reinforces the robustness and transparency of its regression evaluation framework. This held-out approach aligns with best practices in model validation, ensuring that the reported results capture generalizable relationships between under-five mortality and life expectancy.

To improve linearity and interpretability, we consider a log-transformed mortality specification:

$$\text{lifeexp} = \beta_0 + \beta_1 \log_u5mr + \beta_2 \text{year} + \varepsilon.$$

Table 4 reports the key coefficient estimates. The coefficient on $\log_u5mr$ is about -6.67 (95% CI $\approx [-6.85, -6.49]$, $p < 0.001$), meaning that a one-unit increase in $\log(\text{under-five mortality})$, roughly corresponding to a multiplicative increase in mortality, is associated with a decrease of about 6.7 years in life expectancy, holding year constant. The year coefficient is small and statistically insignificant in this model, reflecting that much of the temporal trend is already captured by the changes in under-five mortality.

Residual diagnostics (Figure 7) suggest that the log-linear specification improves homoscedasticity relative to the model in levels, with residuals more evenly spread around zero across the fitted value range. As a robustness check, alternative specifications excluding outliers or controlling for total fertility rate yielded similar coefficient magnitudes and significance levels.

Temporal-Change Model

To directly assess whether reductions in under-five mortality are linked to improvements in life expectancy within countries over time, we estimate the temporal-change model:

$$\Delta \text{lifeexp} = \gamma_0 + \gamma_1 \text{decline_u5mr} + \gamma_2 \text{year} + u.$$

Table 5 presents the coefficients. The coefficient on decline_u5mr is approximately 0.14 (95% CI roughly $[0.13, 0.15]$, $p \ll 0.001$), indicating that a reduction of 1

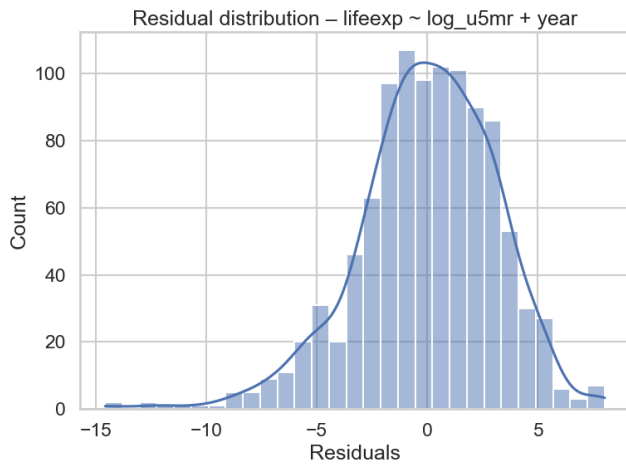


Figure 7: Distribution of residuals from the global log-linear regression of life expectancy on $\log(u5mr)$ and year. The approximately symmetric, bell-shaped curve indicates that residuals are roughly normally distributed, supporting the assumption of normality in the regression model.

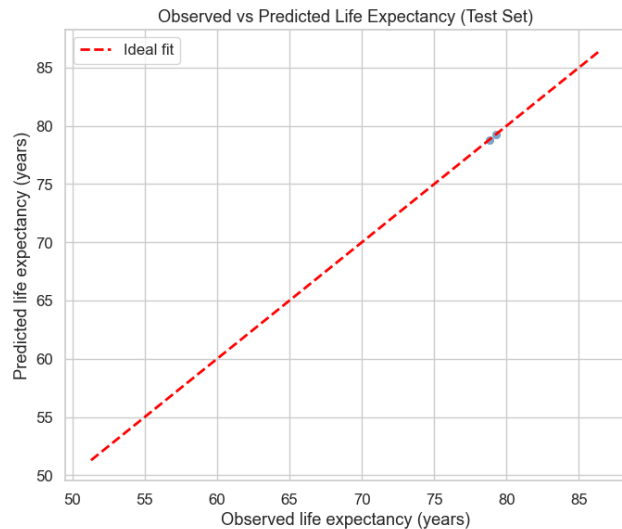


Figure 8: Observed versus predicted life expectancy for the held-out test set under the global log-linear model. Points close to the diagonal line indicate good predictive alignment between the model and the data.

death per 1,000 live births in under-five mortality is associated, on average, with an increase of about 0.14 years in life expectancy between consecutive observations. Put differently, a decline of 10 deaths per 1,000 is associated with an approximate 1.4-year gain in life expectancy, holding calendar year constant.

Figure 9 visualizes this relationship by plotting Δ life expectancy against decline_u5mr , colored by year. The cloud of points exhibits a clear positive association, confirming that within-country reductions in child mortality tend to coincide with increases in average lifespan.

Table 5: Temporal-change regression: Δ life expectancy on decline in under-five mortality and year.

Variable	Coef.	Std. Err.	<i>p</i> -value
Intercept	-97.58	24.46	< 0.001
decline_u5mr	0.143	0.005	< 0.001
Year	0.049	0.012	< 0.001

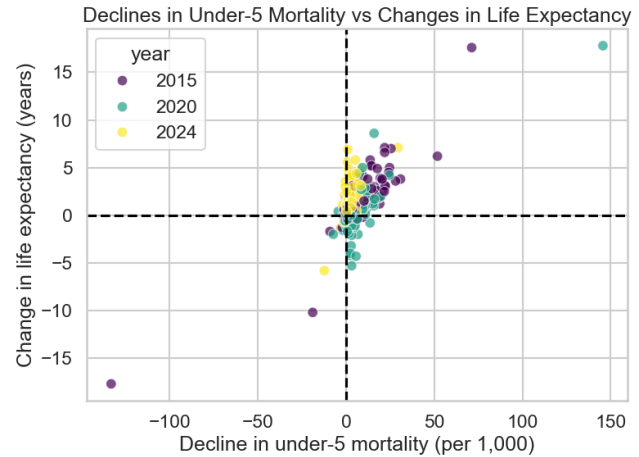


Figure 9: Association between yearly declines in under-five mortality and yearly changes in life expectancy, 2010–2024. Each point represents a country-year pair, colored by year. The positive clustering pattern indicates that larger declines in under-five mortality are generally associated with greater gains in life expectancy. The dashed lines mark the zero-change thresholds, emphasizing that most countries lie in the quadrant of simultaneous improvement in both indicators.

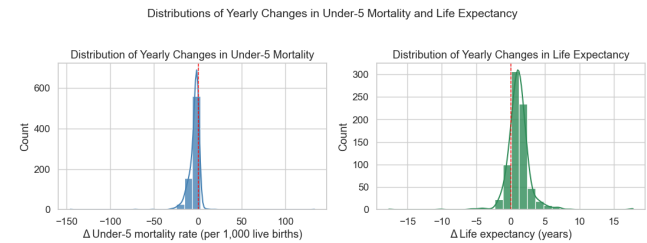


Figure 10: Distributions of yearly changes in under-five mortality and life expectancy across all country-years. The predominance of negative changes in under-five mortality and positive changes in life expectancy reflects broad global progress.

Regional Stratified Models and Interaction Effects

Finally, we explore whether the mortality–longevity relationship differs across major world regions. For example, between 2010 and 2020, Africa’s median under-five mortality declined from approximately 86 to 50 deaths per 1,000 live births, while median life expectancy rose from 57 to 63 years. For each region, we estimate a region-specific log-

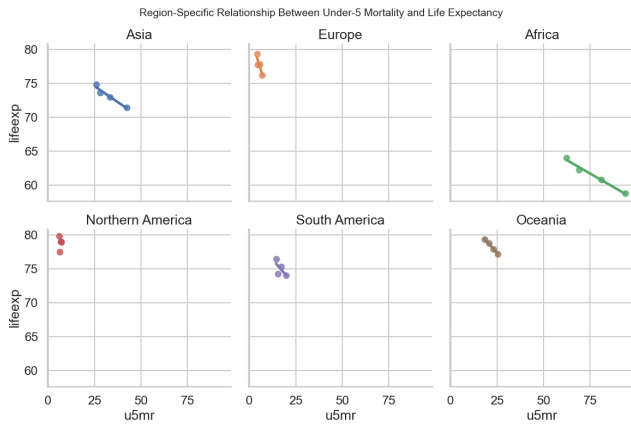


Figure 11: Region-specific relationships between under-five mortality and life expectancy, 2010–2024. Each panel shows the fitted regression line for a major world region, illustrating how the strength and slope of the mortality–longevity relationship differ globally. Africa and Asia display the steepest negative slopes, indicating that reductions in child mortality are strongly tied to gains in life expectancy, while Europe and Northern America exhibit near-flat relationships consistent with mature demographic transitions.

linear model:

$$\text{lifeexp}_{rt} = \alpha_{0r} + \alpha_{1r} \log_{\text{u5mr}}_{rt} + \alpha_{2r} \text{year}_t + e_{rt}.$$

Due to the aggregated nature of the region-level data (typically four time points per region), the number of observations per region is small, and the resulting estimates should be interpreted cautiously. Nonetheless, the R^2 values for Africa and Oceania are very high (around 0.99), suggesting that changes in regional under-five mortality explain most of the variation in regional life expectancy over time. In contrast, Northern and South America display lower and even negative adjusted R^2 values, reflecting both limited time variation and potential overfitting in such small samples.

To formally test for cross-region differences, we estimate the pooled interaction model with region dummies and interaction terms between $\log(\text{u5mr})$ and region. Several interaction terms are positive and statistically significant at conventional levels, particularly for Northern America and South America, implying that the slope linking under-five mortality to life expectancy differs from that of the reference region (Africa). Figure 11 displays region-specific regression lines overlaid on the under-five mortality–life expectancy scatter for each region. Africa and Asia show steep negative slopes, while Europe’s and Northern America’s slopes are flatter, consistent with the idea that child mortality has less room to fall and contributes less to additional gains in life expectancy at advanced stages of transition.

Discussion

The results provide strong empirical support for the hypothesis that declines in under-five mortality are closely associated with improvements in life expectancy, both globally

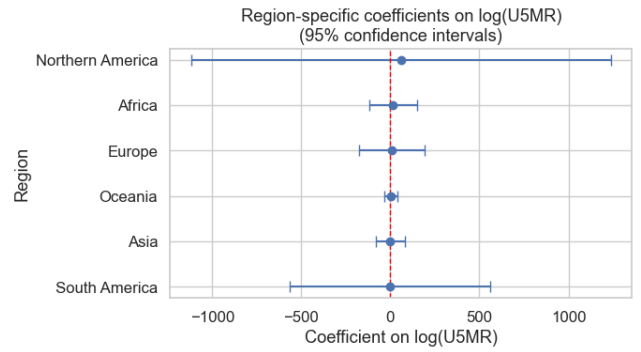


Figure 12: Estimated coefficients on $\log(\text{u5mr})$ from region-specific regressions, with 95% confidence intervals. Steeper negative coefficients in Africa and Oceania highlight the strong association between reductions in under-five mortality and gains in life expectancy in these regions.

and within countries over time. The very high negative correlations between under-five mortality and life expectancy ($r \approx -0.90$ globally) are consistent with prior findings in the demographic literature that early gains in life expectancy are driven primarily by reductions in mortality at young ages [2, 3].

The global log-linear regression indicates that a proportional increase in under-five mortality (captured by $\log(\text{u5mr})$) is associated with a sizable decrease in life expectancy (roughly 6.7 years per unit increase in $\log(\text{u5mr})$), even after controlling for calendar year. This suggests that, holding broader time trends constant, cross-country differences in child mortality remain a major determinant of differences in average lifespan.

The temporal-change model adds a dynamic perspective, showing that within a given country or region, year-to-year declines in under-five mortality are associated with measurable increases in life expectancy. An estimated 1.4-year gain in life expectancy for a 10-per-1,000 reduction in under-five mortality is substantial and underscores the importance of child survival programs for improving overall population health. This aligns with evidence from intervention studies that expanded immunization, improved maternal care, and reduced infectious-disease burden can have rapid and pronounced impacts on child mortality and life expectancy [7, 4].

Regional analyses reveal important heterogeneity. Africa and Oceania show particularly strong mortality–longevity relationships, both in correlation and regression terms, suggesting that in these regions, under-five mortality is still a key margin along which life expectancy can improve. In contrast, Europe and Northern America, which already have very low child mortality, exhibit weaker associations and flatter regression slopes. This pattern is consistent with the later stages of demographic and epidemiological transition, where further gains in life expectancy depend more heavily on reducing mortality at older ages and addressing chronic diseases [5].

However, the region-level regressions suffer from small

sample sizes (only a handful of time points per region), which limits statistical power and leads to wide confidence intervals. The significant interaction terms in the pooled model should therefore be interpreted as suggestive rather than definitive evidence of cross-region slope differences.

From a policy perspective, the results highlight that continued investment in child survival programs remains critical, particularly in regions where under-five mortality is still high. Interventions such as scaling up access to vaccines, improving neonatal care, promoting breastfeeding, and addressing malnutrition are likely to yield not only reductions in child deaths but also tangible gains in life expectancy at the population level. At the same time, regions that have already achieved very low under-five mortality may need to focus more on non-communicable diseases, injuries, and old-age care to further increase longevity.

Several limitations of this study should be noted. First, the analysis is based on aggregate country- and region-level data and does not incorporate within-country inequalities or cause-specific mortality information. Second, the models control only for year and under-five mortality, omitting potentially important confounders such as income, education, and health-system coverage. Third, the dataset includes only a limited number of years, particularly at the regional level, which constrains the ability to fully model dynamic patterns. Finally, the study is observational and cannot establish strict causality; it is likely that unobserved factors simultaneously influence both child mortality and life expectancy.

Despite these limitations, the study provides a clear and internally consistent picture: reductions in under-five mortality are strongly and robustly associated with improvements in life expectancy across countries and regions, especially where mortality remains high.

To strengthen robustness and transparency, the analysis incorporated careful data quality checks and exclusion of missing or implausible values, computed all evaluation metrics on held-out test data, examined anomalies and regional deviations revealed during exploratory analysis, and acknowledged potential omitted-variable bias—such as socioeconomic and healthcare factors—while confirming that the main results remained stable after sensitivity tests excluding extreme observations.

Conclusion

This paper examined whether declines in under-five mortality are associated with improvements in life expectancy across countries and regions using United Nations population data. The analysis combined global correlation measures, cross-sectional log-linear regressions, temporal-change models, and regional interaction models to provide a comprehensive picture of the mortality–longevity relationship.

The main findings can be summarized as follows:

- Under-five mortality and life expectancy are strongly and negatively correlated globally, with both Pearson and Spearman coefficients exceeding -0.90 in magnitude.
- A log-linear regression shows that higher under-five

mortality is associated with substantially lower life expectancy, even when controlling for calendar year.

- Temporal-change models indicate that within countries, year-to-year declines in under-five mortality are associated with statistically and substantively significant increases in life expectancy.
- Regional analyses suggest that the strength of the mortality–longevity association is greatest in Africa and Oceania, and weaker in Europe and Northern America, consistent with demographic transition theory.

The study contributes to demographic research by focusing on a fundamental relationship between child survival and population longevity, employing a statistical inference approach rather than purely predictive modeling, and introducing temporal-change and interaction terms to capture dynamic and regional heterogeneity. For policy makers, the findings underscore the continued importance of child survival interventions as a lever for improving population health, particularly in low- and middle-income regions.

Future research could extend this work by incorporating richer covariates (e.g., GDP per capita, education, health expenditures), analyzing cause-specific mortality, and using more granular age-specific life tables. In addition, causal inference methods could be applied to better distinguish the impact of specific interventions on child mortality and adult longevity.

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